

Thermodynamics

Grau en Enginyeria Matemàtica i Física

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3. Second Law.

3.1. Heat Engines

Introduction

What are heat engines?



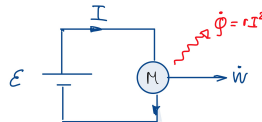
3.1. Heat Engines

Introduction

What are heat engines?



The electric motor is not a heat engine: it uses electrical energy to convert it into work. The heat dissipated can be virtually reduced to virtually zero



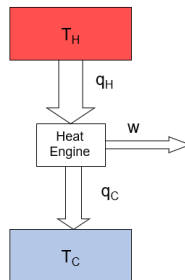
3.1. Heat Engines

Introduction

What are heat engines?



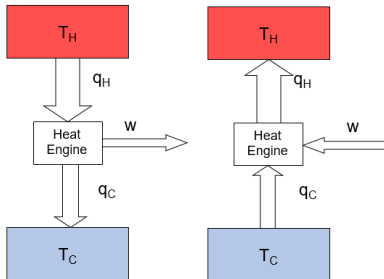
The Stirling engine is a heat engine because it uses heat from the hot spot as the driving element



3.1. Heat Engines

introduction

A heat engine is a system that works cyclically converting heat into work, or vice versa, according to the following scheme:



If, as a result, we get work, we call it **engine**. If the effect is to absorb heat from a cold source and send it to a hot source, we call it a **refrigerator** (if we are interested in the cold we generate at the cold source) or **heat pump** (if we are interested in the heat we send to the hot source)

3.1. Heat Engines

Carnot cycle

Nicolas Léonard Sadi Carnot (1796-1832) was a French *Polytechnicien* engineer who became interested in heat engines.

His treatise *Réflexions sur la puissance motrice du feu et sur les machines propres à développer cette puissance* was fundamental in establishing the existence of entropy as a state function.



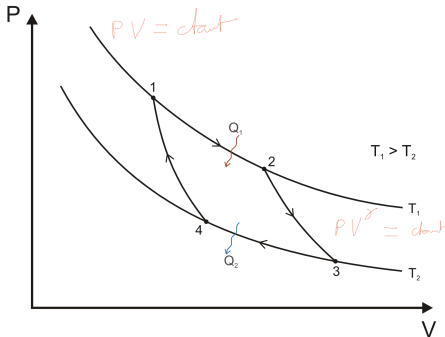
His mental illness and his premature death, however, prevented him from achieving this result.

Entropy, as a state variable, was introduced by Rudolf Clausius (1822-1888) in the article *Über die bewegende Kraft der Wärme und...* ("On the Moving Force of Heat and the Laws of Heat which may be Deduced Therefrom")

3.1. Heat Engines

Carnot cycle

The Carnot cycle is defined as a *reversible* cycle that operates between two temperatures, through isothermal expansions/compressions and adiabatic expansions/compressions. It looks roughly like in the graph,



For ideal gases, the lines satisfy

$$PV = \text{constant (isotherm)}$$

$$PV^\gamma = \text{constant (adiabatic)}$$

3.1. Heat Engines

Carnot cycle

We calculate the heat and work done in each of these stages.

1. Process 1-2: Isothermal expansion. According to what we had seen,

$$W_{12} = -NRT_1 \ln \frac{V_2}{V_1} < 0$$

Heat is obtained from the change in energy, in accordance with the First Law,

$$Q_{12} = U_2 - U_1 - W_{12} = -W_{12} > 0$$

(for an ideal gas, $U = U(T)$)

2. Process 2-3: Adiabatic expansion: $Q_{23} = 0$. We calculated the work earlier.

In this process, we pass from the isotherm T_1 to the T_2 . The connection with the volume variation is as follows:

$$\frac{T_2}{T_1} = \left(\frac{V_2}{V_3} \right)^{\gamma-1}$$

3.1. Heat Engines

Carnot cycle

Similarly,

1. Process 3-4: Isothermal compression.

$$W_{34} = -NRT_2 \ln \frac{V_4}{V_3} > 0$$

In accordance with the First Law,

$$Q_{34} = U_4 - U_3 - W_{34} = -W_{34} < 0$$

2. Process 4-1: Adiabatic compression: $Q_{41} = 0$.

In this process, we go from the isotherm T_2 to T_1 . The connection with the change in volume is as follows:

$$\frac{T_1}{T_2} = \left(\frac{V_4}{V_1} \right)^{\gamma-1} \Rightarrow \boxed{\frac{V_4}{V_1} = \frac{V_3}{V_2}}$$

3.1. Heat Engines

Carnot cycle

Efficiency

The total work done by the cycle can be obtained from the First Law:

$$\Delta U = 0 = W_{12} + W_{23} + W_{34} + W_{41} + Q_{12} + 0 + Q_{34} + 0$$

That is,

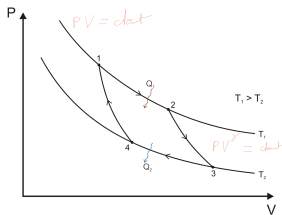
$$W = -Q_{12} - Q_{34} = -|Q_{12}| + |Q_{34}| < 0$$

The **efficiency** is defined as

$$\eta \equiv \frac{|W|}{|Q_{12}|} = \frac{|Q_{12}| - |Q_{34}|}{|Q_{12}|} = 1 - \frac{|Q_{34}|}{|Q_{12}|} = 1 - \frac{NRT_2 \ln \frac{V_3}{V_4}}{NRT_1 \ln \frac{V_2}{V_1}}$$

Since $V_4/V_3 = V_1/V_2$, we will have that

$$\eta = 1 - \frac{T_2}{T_1} \quad (1)$$



3.1. Heat Engines

Carnot cycle

Efficiency

In the case of an engine, we have seen that the efficiency is defined as

$$\eta \equiv \frac{|W|}{|Q_H|}$$

where $|Q_H|$ is the heat absorbed by the cycle coming from the hot source. If the engine is reversible, then we will have

$$\eta_{Carnot} = 1 - \frac{T_C}{T_H}$$

where T_C is the temperature of the cold source (often the ambient temperature) and T_H is the temperature of the hot source.

In the case of a refrigerator, the relevant quantity to define the efficiency, called **coefficient of performance**, is

$$\varepsilon \equiv \frac{|Q_C|}{|W|}; \quad \varepsilon_{Carnot} = \frac{T_C}{T_H - T_C}$$

3.1. Heat Engines

Carnot cycle

Considerations about the Carnot cycle

- ▶ Although we have done the calculation from the ideal gas equations, the result applies generally to any system described by an arbitrary equation of state.
- ▶ The performance of the Carnot engine is the **maximum that can be obtained from any reversible engine**.

This, applied to our Stirling, working between a temperature of 100°C and 0°C, the efficiency is

$$\eta < 1 - \frac{273.15}{373.15} = 0.27$$

- ▶ Every heat engine operating in an irreversible way, has a lower performance.
- ▶ The efficiency of an electric motor can be close to 100%. This is possible because it works by transforming electrical energy into work and not heat into work.
- ▶ Every reversible cycle can be decomposed as a succession of infinitesimal Carnot cycles, and therefore the result can be generalized to any reversible cycle.

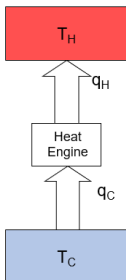
3.1. Heat Engines

Statement of the Second Law

The Clausius statement

As a fact of experience, no engine can be devised which, **working in a cycle**, will produce no effect other than the transfer of heat from a cold source to a hot one.

This is impossible



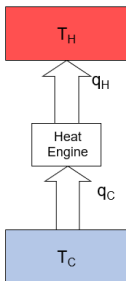
3.1. Heat Engines

Statement of the Second Law

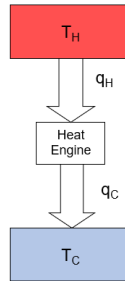
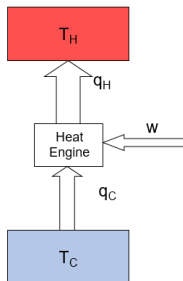
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This, on the other hand, is indeed possible



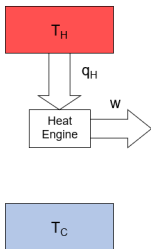
3.1. Heat Engines

Statement of the Second Law

The Kelvin-Planck statement

As a fact of experience, no engine can be devised which, **working in a cycle**, will produce no effect other than taking heat from a source and converting it into work.

This is impossible



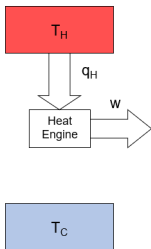
3.1. Heat Engines

Statement of the Second Law

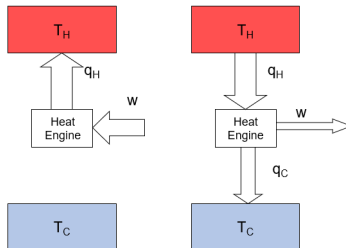
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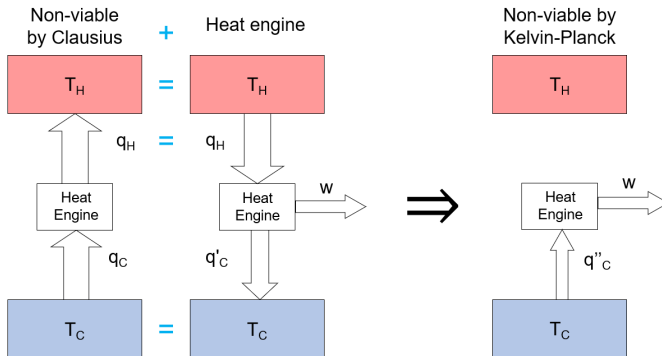


3.1. Heat Engines

Statement of the Second Law

Clausius Kelvin-Planck equivalence

Clausius implies Kelvin-Planck. To prove this, we show that the combination of a non-viable engine according to Clausius with a suitable heat engine, as shown below, becomes a non-viable engine according to Kelvin-Planck.

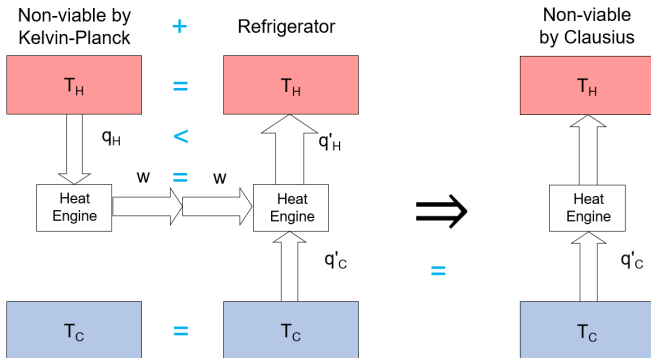


3.1. Heat Engines

Statement of the Second Law

Clausius Kelvin-Planck equivalence

Kelvin-Planck implies Clausius. To prove this, we show that the combination of a non-viable engine according to Kelvin-Planck with a suitable refrigerator, as shown below, becomes a non-viable engine according to Clausius.



3.2. Entropy

Definition of entropy

Finally, Clausius recognized that when analyzing the Carnot cycle, it had a hidden surprise:

Indeed, if we calculate the quotient between the heat entering Q_{12} and the heat leaving the cycle, Q_{34} , we obtain

$$\frac{Q_{34}}{Q_{12}} = \frac{NRT_2 \ln(V_4/V_3)}{NRT_1 \ln(V_2/V_1)} = -\frac{T_2}{T_1}; \text{ recall that } \boxed{\frac{V_4}{V_1} = \frac{V_3}{V_2}}$$

Then,

$$\boxed{\frac{Q_{34}}{T_2} + \frac{Q_{12}}{T_1} = 0} \quad (2)$$

Conclusion

The quantity Q/T , summed over a reversible cycle, is zero. This suggests that this quantity is a **new state function**, which Clausius named **Entropy**, which literally means *evolution*, since its growth indicates the natural sense of the evolution of processes that happen spontaneously.

3.2. Entropy

Definition of entropy

Final considerations about entropy

- ▶ Since any reversible cycle (for any system) can be decomposed into differential Carnot cycles, we can write, in general

$$\oint \frac{dQ^{rev}}{T} = 0$$

- ▶ We can then define the entropy from the heat transferred **reversibly with the source at temperature T** ,

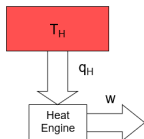
$$dS \equiv \frac{dQ^{rev}}{T} \quad (3)$$

3.2. Entropy

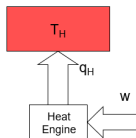
Clausius' inequality

Let's consider the operation of these systems:

Non-viable engine



Heater



Since the first engine is not feasible, the only possibility is that Q_H be necessarily negative, $Q_H < 0$. If the heater were reversible, from equation (2), we would reach the conclusion that,

$$\frac{Q_H^{rev}}{T_H} = 0 \quad (4)$$

as there is only heat transfer to the hot source. Therefore, we conclude that

$$\frac{Q_H}{T_H} < \frac{Q_H^{rev}}{T_H} = \Delta S$$

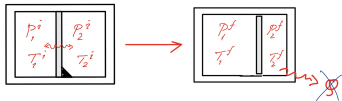
This led Clausius to state that, when there are irreversible processes,

$$dS > \frac{dQ}{T_{ext}} \quad (5)$$

3.2. Entropy

Maximum entropy

So what happens in a thermally isolated system that breaks free from its internal constraints?



Then,

$$dS > \frac{dQ}{T_{ext}} = 0$$

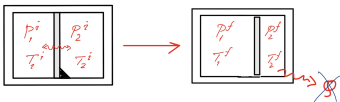
$$dS > 0$$

(6)

3.2. Entropy

Maximum entropy

So what happens in a thermally isolated system that breaks free from its internal constraints?



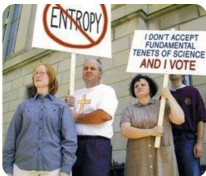
Then,

$$dS > \frac{dQ}{T_{\text{ext}}} = 0$$

$$dS > 0$$

(6)

- The entropy of the universe (as an isolated system) does not stop growing.

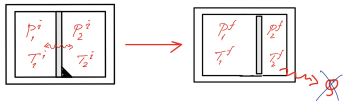


It was thought that the continuous increase in entropy would lead to the thermal death of the universe... Some disagree.

3.2. Entropy

Maximum entropy

So what happens in a thermally isolated system that breaks free from its internal constraints?



Then,

$$dS > \frac{dQ}{T_{\text{ext}}} = 0$$

$$dS > 0$$

(6)

- ▶ The entropy of the universe (as an isolated system) does not stop growing.
- ▶ **Principle of maximum entropy:** when an isolated system is freed from its internal bonds, the system evolves until its entropy reaches its **maximum**.

3.2. Entropy

Computation of entropy changes

To calculate the entropy change between two states, it is often overlooked that one can make use of the equations of state of the system in question to determine the entropy at each state. However, we often do not know the form of the equation of state that provides us with this information.

In this case, we will use the definition of entropy change in terms of the heat transferred **reversibly** between the initial and final states.

It is very important to realize that we can only do this calculation through a reversible transformation, which can be imagined and not coincide with the real physical process that has taken place, as only the initial and final states must coincide, because entropy is a state function.

Next, we show a few examples.

3.2. Entropy

Computation of entropy changes

Reversible heat transfer

Just as for mechanical work the external pressure was the control variable, for heat transfer it is the external temperature.

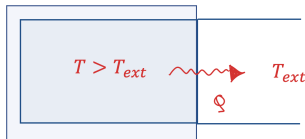
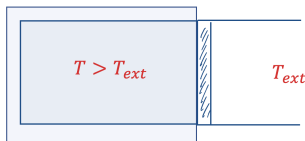
3.2. Entropy

Computation of entropy changes

Reversible heat transfer

Just as for mechanical work the external pressure was the control variable, for heat transfer it is the external temperature.

If we have an isolated system at a temperature $T > T_{ext}$, when we put it in thermal contact with the outside, the heat flows irreversibly



Procés IRREVERSIBLE

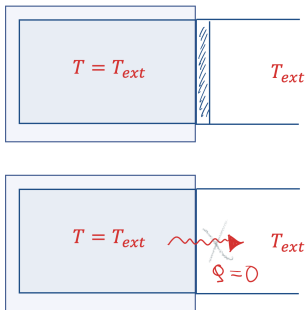
3.2. Entropy

Computation of entropy changes

Reversible heat transfer

Just as for mechanical work the external pressure was the control variable, for heat transfer it is the external temperature.

If we increase the external temperature until $T = T_{ext}$ and remove the insulating wall, leaving the system in thermal contact with the outside, there will be no change, thus $Q = 0$. If we slowly decrease T_{ext} , we will have a reversible flow of heat from the inside to the outside. Conversely, if we raise T_{ext} , the heat will flow in the opposite direction.



Procés REVERSIBLE.

En treure la paret adiabàtica no flueix calor perquè $T_{ext} = T$.

En disminuir la temperatura externa, la calor flueix. El procés es pot invertir simplement augmentant la temperatura externa.

3.2. Entropy

Computation of entropy changes

Reversible heat transfer

Example 3.2 in Levine

EJEMPLO 3.2 ΔS para calentamiento con P constante

La capacidad calorífica específica c_p del agua es casi constante a $1.00 \text{ cal}/(\text{g}^\circ\text{C})$ en el intervalo de temperatura de 25 a 75°C a 1 atm (figura 2.15). *a)* Determine ΔS cuando 100 g de agua se calientan reversiblemente de 25 a 50°C a 1 atm . *b)* Sin efectuar el cálculo, indique si ΔS para el calentamiento de 100 g de agua de 50 a 75°C a 1 atm será mayor, igual o menor que ΔS para el calentamiento de 25 a 50°C .

a) La capacidad calorífica del sistema es $C_p = mc_p = (100 \text{ g})[1.00 \text{ cal}/(\text{g}^\circ\text{C})] = 100 \text{ cal/K}$. (Un cambio de temperatura de un grado Celsius es igual a un cambio de un kelvin.) Para el proceso de calentamiento, (3.28) con C_p constante produce

$$\begin{aligned}\Delta S &= \int_{T_1}^{T_2} \frac{dq_{\text{rev}}}{T} = \int_{T_1}^{T_2} \frac{C_p}{T} dT = C_p \ln \frac{T_2}{T_1} \\ &= (100 \text{ cal/K}) \ln \frac{323 \text{ K}}{298 \text{ K}} = 8.06 \text{ cal/K} = 33.7 \text{ J/K}\end{aligned}$$

Note that, although the heat transfer is reversible, the system undergoes a finite entropy change. Note, however, that reversibility indicates that the **global entropy change** is zero. To demonstrate this, we calculate the entropy variation of the environment. For the environment, if the system gives off heat, the environment absorbs it in the same amount. We will only have to change the sign, $-dq_{\text{rev}}$. Also, since the temperature of the environment and that of the system are always the same, $dS_{\text{environment}} = -dq_{\text{rev}}/T$. Then, $\Delta S_{\text{environment}} = -\Delta S$.

Therefore, the overall entropy balance is zero for a reversible process.

3.2. Entropy

Computation of entropy changes

Entropy change between two equilibrium states of an ideal gas

We have to **design a reversible process** between the two states. We will consider that we will do a) an isothermal expansion-compression at T_1 between V_1 and V_2 , followed by b) a transfer-absorption of heat between T_1 and T_2 at constant V .

- The entropy change during the first process will be

$$\Delta S_a = \int \frac{dQ_{rev}}{T} = \frac{Q_{rev}}{T_1} = -\frac{W_{rev}}{T_1} = NR \ln \frac{V_2}{V_1}$$

- The second process is simply a reversible transfer or absorption of heat at constant V

$$\Delta S_b = \int \frac{dQ_{rev}}{T} = \int_{T_1}^{T_2} dT \frac{C_V(T)}{T}$$

Therefore,

$$\Delta S = \Delta S_a + \Delta S_b = NR \ln \frac{V_2}{V_1} + \int_{T_1}^{T_2} dT \frac{C_V(T)}{T} \quad (7)$$

For a monoatomic ideal gas

$$\Delta S = NR \ln \frac{V_2}{V_1} + \frac{3}{2} NR \ln \frac{T_2}{T_1} \quad (8)$$

3.2. Entropy

Computation of entropy changes

Entropy change in an adiabatic expansion of an ideal gas against a constant pressure

Example 3.3 in Levine

EJEMPLO 3.3 ΔS para la expansión en el vacío

Considere que n moles de un gas perfecto experimentan una expansión libre adiabática en el vacío (el experimento de Joule). *a)* Exprese ΔS en términos de las temperaturas y volúmenes iniciales y finales. *b)* Calcule ΔS_m si $V_2 = 2V_1$.

a) El estado inicial es T_1 , V_1 , y el final, T_1 , V_2 , donde $V_2 > V_1$. T es constante porque $\mu_J = (\partial T / \partial V)_U$ es cero en un gas perfecto. Aunque el proceso es adiabático ($q = 0$), ΔS no es cero porque el proceso es irreversible. La ecuación (3.30) produce $\Delta S = nR \ln(V_2/V_1)$, pues la integral de temperatura en (3.30) es cero cuando $T_2 = T_1$.

b) Si el recipiente original y el recipiente evacuado son de igual volumen, entonces $V_2 = 2V_1$ y $\Delta S = nR \ln 2$. Se tiene

$$\Delta S/n = \Delta S_m = R \ln 2 = [8.314 \text{ J/(mol K)}](0.693) = 5.76 \text{ J/(mol K)}$$

If we know the equation of state of the gas, we can know the final temperature. In this case, being an ideal gas, the internal energy depends only on the temperature. We assume that it is a monatomic gas, $U(T) = \frac{3}{2}NRT$. Then,

$$T_2 = T_1 - \frac{P_{\text{ex}}}{\frac{3}{2}NR}(V_2 - V_1) < T_1$$

This completely fixes the final state the gas will reach, as a function of V_2 , P_{ex} , and initial state variables V_1 and T_1 . Then, this expression for T_2 can be substituted into the expression for ΔS of a monoatomic ideal gas, equation (8).

Assume that the gas expands against a constant external pressure P_{ex} . We can directly calculate the change in the internal energy of the gas

$$\Delta U = W = -P_{\text{ex}}(V_2 - V_1)$$

We will assume that the final pressure of the gas is the external pressure. We also know the initial temperature T_1 .

3.2. Entropy

Computation of entropy changes

Entropy change in an adiabatic expansion against a vacuum

If the external pressure is zero, the work is zero and the change in internal energy is also zero. For the ideal gas, this also implies that the temperature will not undergo any variation, and we will have the expression in the example (see also below).

Note that in this case, the system does not interact with the environment, so both the work (as $P_{\text{ext}} = 0$) and the heat (adiabatic walls) are zero. However, the entropy change is positive, as $V_2 > V_1$ (see equation (7)):

$$\Delta S = NR \ln \frac{V_2}{V_1} > 0$$

This is the key to the distinction between reversible and irreversible processes: Since the heat is zero, one might think that the entropy change is zero. However, due to the *irreversible* nature of this problem, the equality does not apply but Clausius's inequality, which states that

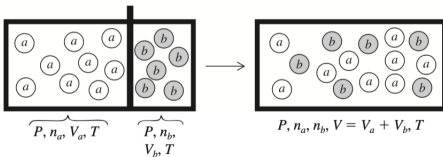
$$\Delta S > \int \frac{dq}{T_{\text{ext}}} = 0 \quad \text{In this case, then } \Delta S > 0$$

3.2. Entropy

Computation of entropy changes

Entropy of mixing

A most relevant case is mixing of components. We can study the increase in entropy due to the mixture of two ideal gases.



When we remove the wall, the system mixes irreversibly. We will assume that the box is closed but the walls are diathermal, with the external temperature T_{ext} fixed. We will also assume that the pressures on both sides of the wall are the same, P , (in order to avoid expansion/compression work when removing the wall), and focus on the pure mixing process. We cannot do the direct entropy calculation because the mixing process is irreversible.

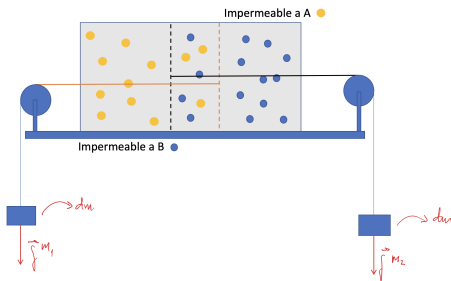
Therefore, we must invent a **reversible process** between the initial and final states, by which we can calculate the heat transferred.

3.2. Entropy

Computation of entropy changes

Entropy of mixing

This process can be thought of as the permeable wall being actually two semi-permeable walls that let one of the substances through, respectively. These walls act like pistons that are connected to the outside where they do reversible work (e.g. lifting a mass that we reduce little by little), as in the figure. Heat interactions are allowed through the diathermic walls of the system.



The process is reversible. In addition, we can calculate the work that each gas is doing from the partial pressure that each of them exerts on the corresponding semipermeable wall. Being the constant external temperature, $T = T_{ext}$ throughout the process. Then,

$$W = -N_A RT \ln \frac{V}{V_A} - N_B RT \ln \frac{V}{V_B}$$

Since they are ideal gases, we know that $U = U(T)$ and, so, $Q = -W$. Therefore, if $\Delta S = Q/T$

$$\Delta S = N_A R \ln \frac{V}{V_A} + N_B R \ln \frac{V}{V_B}$$

Since the initial and final pressures are the same, $V = (N_A + N_B)RT/P$, $V_A = N_A RT/P$ and $V_B = N_B RT/P$ and we can arrive at the known expression

$$\Delta S = -RN \left(\frac{N_A}{N} \ln \frac{N_A}{N} + \frac{N_B}{N} \ln \frac{N_B}{N} \right) = -RN (x_A \ln x_A + x_B \ln x_B) > 0$$

Recommended reading

- [1] I. Levine, *Principios de Fisicoquímica*, McGraw-Hill; Chap. 3
- [2] H.B. Callen, *Termodinámica*, Ediciones AC; Chap. 1
- [3] https://ca.wikipedia.org/wiki/Nicolas_L%C3%A9onard_Sadi_Carnot
- [4] https://en.wikipedia.org/wiki/Nicolas_L%C3%A9onard_Sadi_Carnot
- [5] https://ca.wikipedia.org/wiki/Rudolf_Clausius
- [6] https://en.wikipedia.org/wiki/Rudolf_Clausius